



UAS Survey Tool Tutorial

"Precision Starts Here" — Created by Gabriel Allit

The UAS Survey Tool

The UAS Survey Tool is a desktop application designed for professionals in geospatial surveying, particularly those using Unmanned Aerial Systems (UAS). The tool facilitates the planning of survey missions by allowing users to define Areas of Interest (AOI) using KMZ/KML files, generate Ground Control Points (GCPs) and verification points, compute flight parameters, and export deliverables for field use. Version 2.0 enhances the user experience with a modern UI, advanced layout modes, and robust logging capabilities.

Purpose:

- Generate GCPs and verification points for survey accuracy using novel computations, terrain analysis, vegetative analysis and other tools to help generate the optimal GCP location.
- Compute optimal flight parameters for UAS missions.
- Export deliverables for use in surveying software (e.g., Google Earth, GIS tools).

App Functions

The UAS Survey Tool provides the following core functions:

1. Loading KMZ/KML Files:

- Users can load a KMZ/KML file to define the AOI.
- The app extracts the polygon and calculates the area in acres.

2. Generating Survey Plans:

- Generates GCPs and verification points based on user-defined parameters (spacing, layout mode, sequence type).
- Computes flight parameters (e.g., altitude, speed, resolution) for the survey mission.

3. Exporting Deliverables:

- Exports survey data in various formats (CSV, KMZ, PDF, PNG) for use in the field or further analysis.



UAS Survey Tool Tutorial

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App Features

Input Parameters Section

- **KMZ/KML File Input:** Browse and load a KMZ/KML file to define the AOI.

DEM/DSM Auto-Download: Options to auto-download Digital Elevation Model (DEM) and Digital Surface Model (DSM) data, with manual selection if disabled. The **DEM/DSM Auto-Download** feature in the UAS Survey Tool allows users to incorporate elevation data into their survey planning process, which is critical for accurate Ground Control Point (GCP) placement and flight parameter calculations, especially in varied terrain. The tool provides options to either automatically download Digital Elevation Model (DEM) and Digital Surface Model (DSM) data from GIS sources or manually select local DEM/DSM files if the auto-download is disabled. Below is a detailed explanation of this feature:

- **Digital Elevation Model (DEM) Auto-Download:**
 - **Description:** When enabled, the app automatically downloads a DEM file for the Area of Interest (AOI) defined by the loaded KMZ/KML file. A DEM represents the bare-earth elevation of the terrain, excluding vegetation, buildings, and other surface features.
 - **How It Works:**
 - Upon loading a KMZ/KML file, if the **Auto-Download DEM** checkbox is enabled (checked by default), the app queries a GIS data source (e.g., USGS National Map, OpenTopography, or a similar service) to retrieve a DEM file covering the AOI.
 - The app uses the geographical bounds of the AOI (latitude and longitude from the KMZ/KML file) to fetch the appropriate DEM data, typically in GeoTIFF format.
 - The downloaded DEM file is stored temporarily and its path is displayed in the **DEM File** input field (e.g., Auto-downloaded DEM: <path>).
 - **Purpose in Survey Planning:**



UAS Survey Tool Tutorial

"Precision Starts Here" — Created by Gabriel Allit

- DEM data provides the baseline elevation of the terrain, which is essential for accurately placing GCPs and verification points at the correct heights.
- It ensures that flight parameters (e.g., altitude, resolution) account for terrain variations, improving the accuracy of the survey.
- **Use Case:** Ideal for users who don't have a local DEM file or need up-to-date elevation data for their AOI. This is particularly useful in remote or unfamiliar areas where terrain data isn't readily available.
- **Advantages:**
 - Saves time by automatically fetching elevation data without user intervention.
 - Ensures the use of consistent, standardized data from reliable GIS sources.
- **Limitations:**
 - Requires an internet connection to download the data.
 - May fail if the GIS service is unavailable or if the AOI is outside the service's coverage area (e.g., DEM auto-download failed in the Status Log).
- **Digital Surface Model (DSM) Auto-Download:**
 - **Description:** When enabled, the app automatically downloads a DSM file for the AOI. A DSM represents the elevation of the earth's surface, including vegetation, buildings, and other objects, providing a more complete picture of the terrain and its features.
 - **How It Works:**
 - Similar to DEM auto-download, if the **Auto-Download DSM** checkbox is enabled (checked by default), the app queries a GIS data source to retrieve a DSM file for the AOI.
 - The DSM file, also typically in GeoTIFF format, is downloaded based on the AOI's geographical bounds and displayed in the **DSM File** input field (e.g., Auto-downloaded DSM: <path>).



UAS Survey Tool Tutorial

"Precision Starts Here" — Created by Gabriel Allit

- **Purpose in Survey Planning:**
 - DSM data includes surface features, which are critical for understanding obstacles (e.g., trees, buildings) that might affect UAS flight paths.
 - It helps adjust flight parameters to ensure safe clearance over obstacles and accurate data collection.
 - Provides additional elevation data for diagnostic reports (e.g., Elevation Range: 100.50m to 150.75m in the SurveyReport.pdf).
- **Use Case:** Useful for surveys in areas with significant surface features (e.g., urban environments, forested regions) where obstacle avoidance is a concern.
- **Advantages:**
 - Automates the retrieval of surface elevation data, reducing manual effort.
 - Enhances safety and accuracy by accounting for above-ground features.
- **Limitations:**
 - Like DEM auto-download, it requires an internet connection and may fail if the GIS service is unavailable.
 - DSM data may be less accurate in dense vegetation or urban areas due to data source limitations.
- **Manual Selection When Auto-Download Is Disabled:**
 - **Description:** If the **Auto-Download DEM** or **Auto-Download DSM** checkboxes are disabled (unchecked), users can manually select local DEM/DSM files to use instead of downloading them.
 - **How It Works:**
 - Uncheck the **Auto-Download DEM** or **Auto-Download DSM** checkbox to enable manual selection.



UAS Survey Tool Tutorial

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- The corresponding **DEM File** or **DSM File** input field and "Browse" button become active.
- Click the "Browse" button to open a file dialog and select a local GeoTIFF file (.tif or .tiff).
- The selected file path is displayed in the input field (e.g., C:/data/dem.tif), and the app uses this file for elevation data.
- **Purpose in Survey Planning:**
 - Allows users to use their own elevation data, which may be more accurate or specific to their AOI (e.g., high-resolution LiDAR data).
 - Provides flexibility for offline use or when GIS data sources are unavailable.
- **Use Case:** Ideal for users with pre-existing DEM/DSM data, or in scenarios where internet access is unavailable (e.g., field surveys in remote areas).
- **Advantages:**
 - Offers control over the elevation data used, allowing for higher accuracy if better data is available.
 - Works offline, ensuring the app remains functional in all environments.
- **Limitations:**
 - Requires users to have access to compatible DEM/DSM files in GeoTIFF format.
 - Manual selection can be time-consuming if many files need to be processed.
- **Integration with Survey Planning:**
 - **GCP Placement:** The DEM/DSM data is used to adjust the elevation of GCPs and verification points, ensuring they are placed at the correct height relative to the terrain (e.g., elevation field in GCPs.csv).



UAS Survey Tool Tutorial

"Precision Starts Here" — Created by Gabriel Allit

- **Flight Parameter Adjustments:** Elevation data informs flight parameters like altitude and speed, ensuring the UAS maintains safe clearance and achieves the desired resolution (e.g., Resolution=4.00cm/pixel).
- **Diagnostics:** The DSM data contributes to the survey report's diagnostics, providing elevation statistics (e.g., Elevation Range, Mean Elevation) and insights into terrain complexity.
- **GCP Spacing:** Select spacing between GCPs (300m, 400m, or 500m).

Layout Mode: Choose from various patterns (grid, triangular, diamond, organic, spiral, fibonacci_spiral, fractal, chaotic). The **Layout Mode** feature in the UAS Survey Tool allows users to define how Ground Control Points (GCPs) and verification points are distributed within the Area of Interest (AOI) defined by the KMZ/KML file. Each pattern offers a unique approach to point placement, catering to different terrain types, survey requirements, and accuracy needs. Below is a detailed explanation of each layout mode:

- **Grid:**
 - **Description:** Places GCPs in a uniform grid pattern, forming a regular lattice of points with equal spacing in both the horizontal and vertical directions.
 - **Characteristics:**
 - Points are arranged in a rectangular grid (e.g., a 5x5 grid if the AOI and spacing allow).
 - Spacing between points is consistent, determined by the user-defined **GCP Spacing** (e.g., 300m).
 - **Use Case:** Ideal for flat, open terrain where uniform coverage is desired. This pattern ensures even distribution and is simple to implement in the field.
 - **Advantages:** Easy to set up and interpret; suitable for areas with minimal elevation changes.
 - **Limitations:** May not adapt well to complex terrain with significant elevation variations or obstacles.
- **Triangular:**
 - **Description:** Arranges GCPs in a triangular lattice, where each point forms an equilateral triangle with its neighbors.
 - **Characteristics:**
 - Points are placed at the vertices of equilateral triangles, creating a hexagonal pattern when viewed as a whole.
 - Provides a slightly denser packing compared to a grid, with points equidistant from their neighbors.



UAS Survey Tool Tutorial

"Precision Starts Here" — Created by Gabriel Allit

- **Use Case:** Suitable for flat to moderately varied terrain where a more compact arrangement is needed. Often used in photogrammetry for better overlap.
- **Advantages:** Offers better coverage efficiency than a grid; points are more evenly spaced in all directions.
- **Limitations:** Slightly more complex to set up in the field compared to a grid; may not adapt well to highly irregular terrain.
- **Diamond:**
 - **Description:** Uses a rotated grid pattern, where the grid is tilted at a 45-degree angle, forming a diamond-shaped lattice.
 - **Characteristics:**
 - Points are arranged in a grid but rotated, creating a diamond pattern.
 - Spacing is still uniform, but the orientation provides a different coverage pattern compared to the standard grid.
 - **Use Case:** Useful for areas where a grid-like structure is desired but with a different alignment, often to align with terrain features (e.g., following a river or road).
 - **Advantages:** Provides a balanced distribution with a unique orientation; can improve coverage in certain scenarios.
 - **Limitations:** Similar to the grid, it may not adapt well to complex terrain; the rotated pattern can be harder to implement in the field.
- **Organic:**
 - **Description:** Randomly places GCPs within the AOI while maintaining a minimum spacing constraint to avoid clustering.
 - **Characteristics:**
 - Points are distributed pseudo-randomly, ensuring they are at least the user-defined spacing apart (e.g., 300m).
 - Uses an algorithm to prevent points from being too close, creating a more natural, scattered distribution.
 - **Use Case:** Ideal for uneven terrain with natural features (e.g., forests, hills) where a regular grid might not fit well.
 - **Advantages:** Adapts to irregular terrain; avoids placing points in inaccessible areas by randomizing placement.
 - **Limitations:** Less predictable placement can make field setup more challenging; coverage may be less uniform.
- **Spiral:**
 - **Description:** Distributes GCPs in a spiral pattern, starting from the center of the AOI and spiraling outward.
 - **Characteristics:**
 - Points follow a logarithmic spiral, with spacing increasing as the spiral expands.



UAS Survey Tool Tutorial

"Precision Starts Here" — Created by Gabriel Allit

- The density of points is higher near the center and decreases outward.
 - **Use Case:** Suitable for circular or near-circular AOIs, or when focusing on a central area with less emphasis on the periphery.
 - **Advantages:** Concentrates points in the center, which can be useful for detailed surveys in a specific area; visually appealing pattern.
 - **Limitations:** Uneven distribution (denser in the center); may not provide adequate coverage for the entire AOI if the spiral doesn't extend far enough.
- **Fibonacci Spiral:**
 - **Description:** Uses a Fibonacci spiral pattern, inspired by natural patterns (e.g., sunflower seeds), to place GCPs.
 - **Characteristics:**
 - Points are placed along a spiral where the angle between successive points follows the golden angle (approximately 137.5 degrees), derived from the Fibonacci sequence.
 - Creates a naturally balanced distribution that mimics organic patterns.
 - **Use Case:** Ideal for surveys requiring a natural, evenly distributed pattern, especially in areas with organic shapes or varied terrain.
 - **Advantages:** Provides a more uniform coverage than a regular spiral; aesthetically and mathematically elegant.
 - **Limitations:** Can be complex to implement in the field; may require more points to cover large AOIs evenly.
- **Fractal:**
 - **Description:** Applies a recursive Fibonacci-based placement to create a fractal-like pattern for GCP distribution.
 - **Characteristics:**
 - Uses a fractal algorithm influenced by the Fibonacci sequence, creating self-similar patterns at different scales.
 - Points are placed recursively, with clusters forming smaller fractal patterns within the AOI.
 - **Use Case:** Best for complex terrain where a hierarchical, recursive pattern can adapt to varying scales of features (e.g., mountainous regions with nested valleys).
 - **Advantages:** Highly adaptive to complex terrain; provides a unique, mathematically interesting distribution.
 - **Limitations:** Complex pattern can be difficult to set up in the field; may result in uneven coverage in some areas.
- **Chaotic:**
 - **Description:** Places GCPs in a pseudo-random, chaotic pattern while maintaining a minimum spacing constraint.
 - **Characteristics:**



UAS Survey Tool Tutorial

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- Uses a chaotic algorithm (possibly based on a fractal score, as seen in `filter_coordinates.py`) to distribute points unpredictably.
- Ensures points are at least the user-defined spacing apart (adjusted by a fractal score), preventing clustering.
- **Use Case:** Ideal for highly irregular terrain where traditional patterns (grid, triangular) fail to adapt, such as rugged landscapes with obstacles.
- **Advantages:** Maximizes adaptability to complex terrain; can achieve higher point counts in challenging areas (e.g., 30–40 GCPs, 5 verification points as targeted).
- **Limitations:** Unpredictable placement can make field setup challenging; requires careful planning to ensure coverage.
- **Sequence Type:** Select mathematical sequences for pattern generation (fibonacci, lucas, tribonacci, pell).
- **Modulus:** Auto-calculated or user-defined modulus for sequence generation.

Flight Parameters Section

- **Desired Accuracy:** Set the target accuracy in millimeters (default: 6mm).
- **Flight Altitude:** Define the altitude in meters (default: 80m).
- **LiDAR FOV:** Set the Field of View in degrees (default: 90deg).
- **Pulse Repetition Rate (PRR):** Fixed at 640 kHz for the RECON-Xt LiDAR.

Additional Features

- **Status Log:** Displays real-time logs of app activities in the UI.
- **Modern UI:** Dark-themed interface with a scrollable layout, optimized for various screen sizes.
- **Logging System:** Logs are written to the console, a file (`uas_survey_tool.log` in `~/UAS_Survey_Tool_Logs`), and the UI's Status Log.

App Deliverables/Exports

The UAS Survey Tool generates the following deliverables:

1. **GCPs.csv**
 - **Description:** A CSV file containing the coordinates and metadata of generated Ground Control Points.



UAS Survey Tool Tutorial

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- **Details:**

- Columns: name, lat, lon, elevation, easting, northing, pattern_score, global_offset (if applicable).
- Example Filename: Syptert_AOI_Phase_4_20250420_GCPs.csv.
- Use Case: Import into GIS software for field surveying.

2. VerificationPoints.csv

- **Description:** A CSV file containing the coordinates and metadata of verification points.

- **Details:**

- Columns: Same as GCPs.csv.
- Example Filename: Syptert_AOI_Phase_4_20250420_VerificationPoints.csv.
- Use Case: Used to validate the accuracy of the survey.

3. SurveyPlan.kmz

- **Description:** A KMZ file containing the survey plan, including GCPs and verification points.

- **Details:**

- Contains KML data with folders for GCPs and Verification Points.
- Each point includes coordinates (lon, lat, elevation) and a description (e.g., Pattern Score: 0.75).
- Example Filename: Syptert_AOI_Phase_4_20250420_SurveyPlan.kmz.
- Use Case: Open in Google Earth for visualization.

4. OverlayMap.png

- **Description:** A PNG image showing an overlay map of the AOI, GCPs, and verification points.

- **Details:**



UAS Survey Tool Tutorial

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- Displays the polygon (AOI), GCPs, and verification points in UTM coordinates.
- Example Filename: Syptert_AOI_Phase_4_20250420_OverlayMap.png.
- Use Case: Visual reference for survey planning.

5. SurveyReport.pdf

- **Description:** A PDF report summarizing the survey plan.
- **Details:**
 - Includes polygon coordinates, GCPs, verification points, spacing, layout mode, area (in acres), elevation data, DSM summary, overlay image, and pattern score.
 - Example Filename:
Syptert_AOI_Phase_4_20250420_SurveyReport.pdf.
 - Use Case: Formal documentation for stakeholders.



UAS Survey Tool Tutorial

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User Guide

Step 1: Launch the App

- Double-click UAS_Survey_Tool.exe in the dist folder to launch the app.
- The UI will display with the Input Parameters, Flight Parameters, and Status Log sections.

Step 2: Load a KMZ/KML File

- In the **Input Parameters** section, click the "Browse" button next to "KMZ/KML File".
- Select a KMZ/KML file (e.g., 03_24_2025 Syptert AOI Phase 4.kmz).
- The app will load the file, calculate the area, and display a message in the Status Log (e.g., Loaded KMZ: <path>, Area: 568.40 acres).
- The Modulus field will be auto-filled based on the area.

Step 3: Configure Parameters

- **Input Parameters:**
 - Adjust **GCP Spacing** (e.g., 300m).
 - Select a **Layout Mode** (e.g., Chaotic).
 - Choose a **Sequence Type** (e.g., Fibonacci).
 - Optionally disable DEM/DSM auto-download and manually select files.
- **Flight Parameters:**
 - Set **Desired Accuracy** (e.g., 6mm).
 - Set **Flight Altitude** (e.g., 80m).



UAS Survey Tool Tutorial

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- Set **LiDAR FOV** (e.g., 90deg).

Step 4: Generate a Plan

- Click the "Generate Plan" button.
- The app will generate GCPs and verification points, compute flight parameters, and display results in the Status Log (e.g., Generated 10 GCPs and 5 verification points, Flight Parameters: Accuracy=6.0mm, Altitude=80.0m, ...).

Step 5: Export Deliverables

- Click the "Export Deliverables" button.
- Choose an export directory (e.g., C:/test).
- The app will export the deliverables (GCPs.csv, VerificationPoints.csv, SurveyPlan.kmz, OverlayMap.png, SurveyReport.pdf) and log the results in the Status Log.

Troubleshooting and FAQs

Common Issues

- **KMZ Loading Fails:**
 - Ensure the KMZ/KML file is valid and contains a polygon.
 - Verify that the app has permission to access the file.
- **No GCPs Generated:**
 - Ensure a KMZ file is loaded before generating a plan.
 - Check the Status Log for error messages (e.g., No polygon or CRS loaded).
- **Icon Not Displaying:**
 - Ensure app_icon.ico is in the project directory.
 - Check the console for debug messages (e.g., Failed to load icon: QIcon is null).

FAQs

- **What file formats are supported for input?**



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- KMZ and KML files for AOI definition; TIFF files for DEM/DSM.
- **Can I customize the layout of GCPs?**
 - Yes, choose from multiple layout modes (grid, chaotic, etc.) and sequence types (fibonacci, etc.).
- **Where are logs stored?**
 - Logs are stored in ~/UAS_Survey_Tool_Logs/uas_survey_tool.log.

System Requirements and Installation

System Requirements

- **Operating System:** Windows 10 or later.
- **RAM:** Minimum 4 GB (8 GB recommended).
- **Storage:** At least 500 MB free space for the executable and exported files.
- **Dependencies:** None (all dependencies are bundled in the .exe).

Installation

- The app is distributed as a single executable (UAS_Survey_Tool.exe).
- Place the .exe in a desired directory (e.g., C:\UAS_Survey_Tool).
- Ensure app_icon.ico is in the same directory as the .exe.
- Double-click the .exe to launch the app.

Additional Information

Version History

- **Version 2.0 (April 20, 2025):**
 - Modern UI with dark theme.
 - Added support for multiple layout modes and sequence types.
 - Improved logging system (console, file, UI).
 - Fixed dependency issues (rasterio, pyogrio).

Contact Details



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- **Support:** Contact the developer at Gabriel.allit1@gmail.com.
- **Feedback:** Submit feedback via the email.

Credits

- **Developed by:** [Gabriel Allit/ Salt19 LLC.]
Support Questions? Contact Gabriel Allit — or just shout into the wind.
- **Surveyor Humor:** Why don't drone surveyors panic? Because they always stay grounded.
- **License Coffeeware License:** As long as you're having a coffee, you can use it. If we meet and this tool helped you — buy me one too. Cheers, - Gabriel Allit